Project Report

Designing a microcontroller-based system for Overvoltage and undervoltage protection for electrical Appliance

Objectives: The objective of this project is to design a microcontroller-based system that protects electrical appliances by disconnecting them during power-grid voltage surges. The design was developed in a low-voltage and high-voltage simulation environment.
The system architecture relies on a central microcontroller; here, I used an Arduino Uno microcontroller for this application.

Implementation Flowchart:

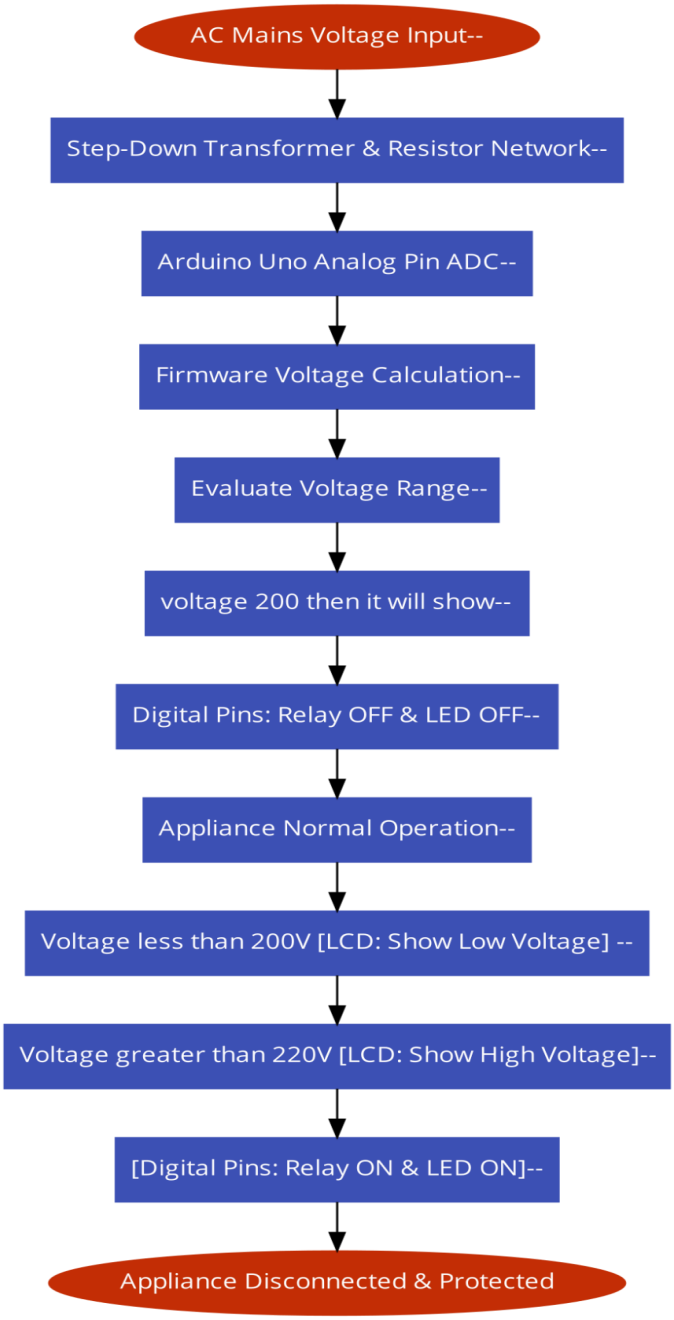


Table: Showing when Relay will trigger

Voltage Condition (AC)	LCD Display Status	Relay State	LED Indicator State	Appliance Status
Below 200V (Low Voltage)	"Low Voltage"	Triggered (Active)	ON	Disconnected (Protected)
200V to 220V (Normal)	"Normal Voltage"	Released (Inactive)	OFF	Connected (Normal Operation)
Above 220V (High Voltage)	"High Voltage"	Triggered (Active)	ON	Disconnected (Protected)